

AI-Driven Complaint Management System : An Automated Framework for Prioritizing, Tracking, and Resolving Public Service Complaints Efficiently

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Abstract—Civic complaint management systems can be inefficient due to unstructured data, manual prioritization, and delayed escalation, resulting in slow response and resolution time. Current platforms for digital engagement and transparency in the real-time civics context still rely on static, rule-based workflows with no dynamic severity assessment and a lack of predictive capability. In an attempt to mitigate these problems, we have proposed an AI-powered scalable framework for intelligent complaint management in this study. It involves feature extraction using NLP for understanding semantics, a multi-task neural network for predicting urgency and time, spatiotemporal clustering for duplicate detection, and workload-aware dynamic routing for optimal department allocation. A mechanism that escalates unresolved cases in hierarchy on a timely basis is also introduced. The application demonstrated substantial improvements with average response time reducing by 81%, resolution time reducing by 68.8%, workload balance index increasing from 0.52 to 0.86, and fake complaint detection achieving 92% accuracy. The results show that the proposed system converts complaints into information for administration to make it a proactive, transparent, and efficient governance process, increasing its accountability, efficiency, and responsiveness to services.

Index Terms—Electricity Fault Reporting, Citizen Participation, Firebase, Machine Learning, Geo-Tagging, Smart City, Digital Governance, Public Utility Management, Fault Prioritization.

I. INTRODUCTION

Public utilities, urban infrastructure, and government services all depend on the efficient handling of civic grievances because it has a direct impact on administrative accountability, transparency, and service efficiency [1]. The majority of current systems, which rely on manual data entry, unstructured complaint processing, and inadequate interdepartmental coordination, are still operationally constrained despite the widespread use of digital grievance platforms [2]. These restrictions result in unclear escalation procedures, slow

resolution times, and inadequate oversight, which eventually erodes public confidence and system efficacy.

With a primary focus on text-based complaint classification, recent research has investigated artificial intelligence-based methods to automate civic grievance handling. An NLP and RNN-based classifier for electricity-sector complaints was proposed by Blandón-Andrade et al. [2], which achieved high classification accuracy but lacked workload-sensitive routing and spatial awareness. An NLP-based categorization system with a moderate level of accuracy was created by Quratul-ain et al. [3], but their method was restricted to textual analysis without clustering or escalation mechanisms. In order to improve workload distribution, Peng et al. [4] introduced a social-sensing framework that used hotline complaint data. However, the system relied on manual workflows and lacked multimodal verification and predictive urgency modeling.

Despite these contributions, current methods treat grievance processing as discrete tasks and do not offer an integrated framework that concurrently supports workload-aware department assignment, urgency prediction, semantic and spatiotemporal clustering, hierarchical escalation, and fake complaint filtering in a scalable and domain-agnostic way [1], [2], [3], and [5]. The shift from reactive complaint handling to proactive, intelligent civic service management is hampered by this lack of a unified architecture.

This paper proposes a generalized AI/ML-driven complaint management architecture that automates grievance classification, prioritization, and resolution across various civic domains in order to close this gap [6], [7], [8]. Three interconnected layers make up the proposed system: (i) an AI-based analysis and prioritization layer that uses neural models and natural language processing to extract semantic information and predict the urgency of complaints [9], [10], [11]; (ii) a decision

and escalation layer that allows workload-aware routing and dynamic departmental assignment for balanced resource utilization [6], [7], [8] and (iii) a visualization and feedback layer that offers real-time dashboards for administrators and citizens to improve transparency and monitoring [12], [13], [14], [6]. The VidyutSaathi case study in the electricity sector illustrates how the architecture is intended to be expandable to other civic services like transportation, water supply, and sanitation [15], [14].

The key contributions of this work include: (i) a multi-task learning model for simultaneous complaint categorization and time prediction; (ii) a spatiotemporal clustering mechanism integrating semantic similarity with density-based algorithms; (iii) a workload-aware assignment model optimizing departmental resource utilization; and (iv) a fake complaint detection module combining NLP filtering, CNN-based image verification, and user credibility scoring.

II. LITERATURE REVIEW

The public and utility sectors have developed digital service delivery from their initial static web-based governance portals to present-day intelligent mobile-first ecosystems that use analytics for service delivery. The first e-governance systems focused on providing transparent government operations while delivering accessible services to citizens through their centralized digital platforms. Yellisetti et al. [13] developed a web-based eligibility system for government programs using PHP and MySQL which showed that centralized data management supports automated benefit verification while making programs more accessible and reducing the need for manual work. This work marked an early step toward digitized citizen–authority interaction and influenced the structured data models adopted in modern utility dashboards.

Researchers began to focus on mobile civic applications which enable real-time citizen-government communication when people started using smartphones. The WASTEAPP Android system [14] developed a household waste management system which used complaint reporting and geo-tagging and Firebase-based synchronization to enable quick complaint resolution. Research on food waste management [10] showed that IoT and cloud automation technologies provide organizations with tools to optimize their resources in real time. The research demonstrated that people could use mobile devices and IoT systems as effective tools which enabled them to participate in civic activities while the study established a system for citizen-officer communication through the government platform.

As the necessity for data had grown, visualization and decision analysis had evolved as integral tools for operational intelligence. Banerjee et al. [11] showed that interactive visualization frameworks enhance situational awareness and accelerate executive decision-making. Farmanbar and Rong [12] developed the Triangulum City Dashboard by implementing their research into urban governance which combines various smart-city indicators into geospatial visualizations. The proposed system aligns with these developments by employing `community_charts_flutter` for interactive heatmaps, regional

analytics, and performance dashboards which support field operations and administrative oversight.

In recent years, analytics has developed into predictive intelligence through the combination of machine learning and cloud orchestration technologies. Aggarwal et al. [6] demonstrated predictive dashboards for resource utilization, while Ahmed and Mueller [16] introduced EnergyScout, a consumer-oriented dashboard for smart-meter analytics. These studies highlight the transition from reactive monitoring to proactive forecasting, which our system extends through predictive ticket-volume analysis and optimized crew dispatch.

The cross-platform frameworks and cloud backends have reached their final state through their technological advancement. Our System uses Flutter to build its user interface and Firebase Firestore to create real-time data synchronization which demonstrates its evolution from basic web governance systems [13] to advanced mobile utility management systems [6] [16] [17] that use predictive analytics to deliver a unified data-based platform for citizens to access statewide electricity distribution services.

III. METHODOLOGY

A. Data Collection and System Overview

More than 20,000 electricity-related complaints were gathered from public records and civic websites and grouped into four levels of severity (E, A1, A2, A3) according to their severity and response time. The Sentence-Transformer model [8] (distiluse-base-multilingual-cased-v1) was employed to produce 512-dimensional multilingual embeddings. These embeddings, together with normalized temporal features and label encodings, were used in a PyTorch-based multi-task learning model to predict the severity of complaints and estimated timelines. The proposed AI-powered Complaint Management System adopts a modular design to efficiently process and manage complaints as depicted in Figure 1. The complaints are first labeled with severity levels and predicted response and resolution times. Spatiotemporal clustering is then performed to detect similar or duplicate complaints based on spatial, temporal, and semantic attributes, allowing severity labeling at the cluster level. The clustered complaints are then directed to a workload-aware assignment module that dynamically reassigns tasks when faced with high workload situations. Finally, the remaining complaints are escalated through the JE-CE hierarchy assisted by dashboard-based monitoring for timely resolution.

B. Algorithmic Design: Working of Each Module

This section details the internal working of each intelligent module in the proposed AI-based architecture.

1) AI-based Complaint Categorization (Urgency Prediction): The AI-based Urgency Prediction module sorts complaints into four severity levels: E, A1, A2, and A3. It also predicts response and resolution times using a multi-task deep learning framework (Figure 2). This module is built as a multi-head neural network PyTorch [18] [19]. The model has

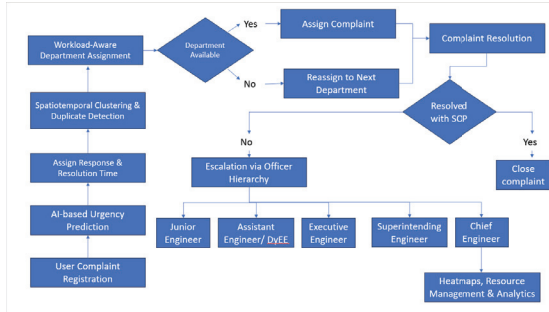


Fig. 1. System Architecture Flowchart

a shared fully connected layer with ReLU activation, which extracts important features from semantic embeddings. Each output head is optimized using categorical cross-entropy loss, and the total loss function is the sum of the three head

$$L = L_{cat} + L_{resp} + L_{resol} \quad (1)$$

The network is trained using the Adam optimizer with a learning rate of 1×10^{-3} for 15 epochs on a GPU to accelerate computation.

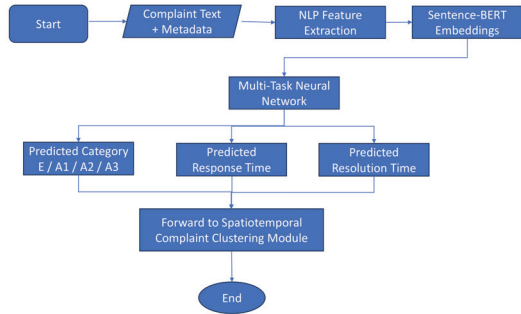


Fig. 2. Pipeline of the AI-based Complaint Categorization Module

2) *Spatiotemporal Complaint Clustering Module (SCCM)*: The Spatiotemporal Complaint Clustering Module (SCCM) cuts down on redundancy by grouping complaints based on their spatial, temporal, and semantic similarities. Each complaint has geolocation data, a timestamp, and text. It also applies spatial normalization, outlier filtering, and keyword-based NLP enrichment. SCCM uses an improved DBSCAN algorithm with spatial thresholds of 100 to 300 meters and temporal thresholds of 24 to 48 hours. It incorporates Sentence-BERT cosine similarity scores above 0.75 for semantic matching. Each cluster receives a severity index (Ψ_c) based on its density and growth rate, which allows for automatic escalation. R-tree indexing in PostGIS achieves $O(N \log N)$ complexity and improves performance by over 70%. This effectively merges duplicates and identifies real issue hotspots.

3) *Workload-Aware Dynamic Department Assignment Module (WDDAM)*: The **Workload-Aware Dynamic Department**

Assignment Module (WDDAM) intelligently allocates clustered complaints to the most suitable department by considering both geographical proximity and real-time workload. Each department D_i is characterized by metrics—capacity (C_i), number of active complaints (N_i), and average resolution time (T_i)—updated dynamically through the administrative dashboard. A *workload index* is computed as:

$$W_i = \alpha \left(\frac{N_i}{C_i} \right) + \beta \left(\frac{T_i}{T_{exp}} \right) \quad (2)$$

Departments with higher W_i values are less preferred for new assignments. An *assignment cost function* is then calculated for each department:

$$A_i = \lambda D_i + (1 - \lambda) W_i \quad (3)$$

where D_i is the normalized Haversine distance and λ (0.6–0.7) controls the trade-off between distance and workload. Complaint is assigned to department with the lowest assignment cost A_i ensuring efficiency and balance. If a department exceeds its workload threshold ($W_i > 0.8$), the system automatically reassigns the complaints. Complaints violating SOP timelines are sent to higher authorities. This helps avoid bottlenecks.

4) *Escalation Mechanism (Hierarchy Control)*: The Escalation mechanism induces timely accountability governed by SOPs for continuous response, resolution and update. If any complaints exceed defined thresholds, they will be forwarded to the next level. This mechanism is useful for timely resolution, accountability and continuous administration.

Escalation follows a hierarchical chain: *Junior Engineer (JE)* $\rightarrow$ *Assistant Engineer (AE) / Deputy Executive Engineer (DyEE)* $\rightarrow$ *Executive Engineer (EE)* $\rightarrow$ *Superintending Engineer (SE)* $\rightarrow$ *Chief Engineer (CE)*

5) *Fake / Redundant Complaint Handling Module*: The module functions in three key stages: First, NLP-based textual validation filters out irrelevant or nonsensical complaints (e.g., greetings, gibberish, or phrases/words that are not even related) while retaining domain-specific terms (e.g., transformer, street-light, power cut, etc.). Second, a CNN-based image-text consistency check flags mismatches or low-confidence cases for human review. Finally, a user credibility scoring system tracks reliability over time, penalizing repeated fake submissions. This pipeline ensures that only genuine complaints proceed to prioritization and resolution, as shown in Figure 3.

IV. RESULTS AND EVALUATION

Evaluation Metrics

The complaint categorization and time prediction models were thoroughly evaluated using classification and regression metrics. Classification performance was measured using precision, recall, F1-score, and accuracy to assess urgency prediction reliability and false alarms. Regression performance was evaluated using MAE, MSE, RMSE, and R^2 to measure the accuracy, robustness, and explanatory power of response and resolution time predictions.

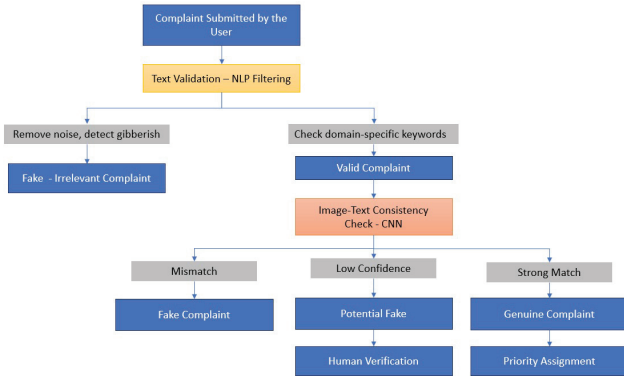


Fig. 3. Automated Complaint Verification Flow

$$\text{Precision} = \frac{TP}{TP + FP} \quad (4)$$

$$\text{Recall} = \frac{TP}{TP + FN} \quad (5)$$

$$\text{F1-Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (6)$$

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (7)$$

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (8)$$

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad (9)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (10)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (11)$$

A. Result

All experiments were implemented using PyTorch and Sentence-BERT on a GPU-enabled Ubuntu system, with evaluation performed using scikit-learn and geospatial processing handled by PostgreSQL/PostGIS.

1) Classification Performance (Complaint Category Prediction):

The classification results in Table I (A) and Figure 4 (A) demonstrate that the proposed AI-based urgency prediction module achieves consistently high performance across all complaint categories. The model attained an overall accuracy of 95%, confirming its effectiveness in distinguishing between critical, high, medium, and low urgency complaints.

Overall, the balanced precision and recall values across all categories indicate that the model does not exhibit bias toward any specific class. This balanced performance ensures reliable

TABLE I
OVERALL PERFORMANCE OF THE PROPOSED SYSTEM

A. Complaint Urgency Classification Performance				
Category / Stage	Precision	Recall	F1-Score	Accuracy / Support
E (Critical)	0.96	0.94	0.95	250
A1 (High)	0.94	0.97	0.95	260
A2 (Medium)	0.93	0.96	0.94	240
A3 (Low)	0.98	0.94	0.96	250
Overall Accuracy				0.95
B. Fake / Redundant Complaint Detection Performance				
Text Validation (NLP)	0.954	0.938	0.946	0.949
Image-Text Consistency (CNN)	0.921	0.907	0.913	0.925
User Credibility Scoring	0.902	0.886	0.894	0.901

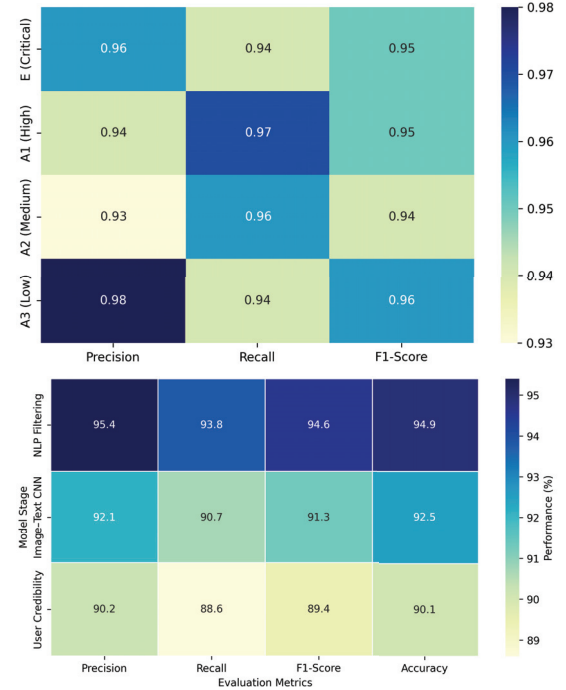


Fig. 4. Performance visualization of (a) complaint urgency classification and (b) fake complaint detection stages.

urgency prediction, enabling faster prioritization, improved decision-making, and efficient resource allocation in real-world complaint management scenarios.

B. Regression Performance (Response and Resolution Time Prediction)

The regression models were used to predict expected response time and resolution time for each complaint. The results, shown in Table II, demonstrate high accuracy and low error values, indicating that predicted times closely match the actual observed times.

TABLE II
REGRESSION METRICS FOR TIME PREDICTIONS

Task	MAE (hr)	MSE (hr ²)	RMSE (hr)	R ² Score
Response Time Prediction	0.42	0.38	0.61	0.90
Resolution Time Prediction	0.37	0.32	0.56	0.92

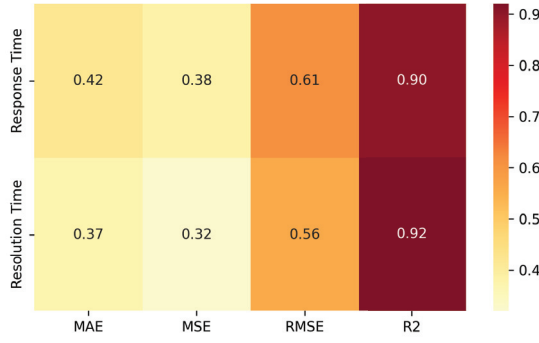


Fig. 5. Precision Heatmap along with model result for complaint categories.

The regression performance of the proposed multi-task model is summarized in Table II and visually represented in Figure 5. The results indicate that the model achieved consistently low error values across both response and resolution time prediction tasks. Specifically, the Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE) values remained below one hour, demonstrating the model's high temporal precision. The R^2 scores of 0.90 for response time and 0.92 for resolution time further confirm the model's strong predictive capability and fit to the observed data.

C. Fake / Redundant Complaint Detection Module

The Fake/Redundant Complaint Handling Module was evaluated on 1,000 labeled complaints containing textual and image data using standard classification and similarity metrics, including Precision, Recall, F1-score, Accuracy, Cosine Similarity, and AUC-ROC. The results are summarized in Table I. The module consists of three sequential stages: textual validation, image-text consistency analysis, and user credibility assessment.

The precision achieved in the text validation stage using the proposed NLP approach is of 95.4%, which indicates successful filtering of irrelevant complaints while preserving authentic ones. The image-text consistency stage achieved a cosine similarity of 0.89 and an AUC-ROC of 0.94, suggesting a high level of textual and visual information, and the accurate detection of misinformation matches. The user credibility scoring stage successfully produced an AUC-ROC of 0.91, which demonstrates an accurate detection of suspicious complaint patterns.

Overall, the consistently high performance across all stages confirms that the proposed module effectively detects fake and redundant complaints before they enter the prioritization pipeline, thereby enhancing operational efficiency, decision accuracy, and system reliability.

D. Comparative Evaluation – Traditional vs. Proposed System

In order to assess the effectiveness and the real-life implications of the AI-driven Complaint Management System, a controlled experimental study was conducted on a college campus. The evaluation compared the created model in combination with the traditional workflow of grievance handling, and

as such, it is representative of existing government complaint platforms including the Centralized Public Grievance Redress and Monitoring System (CPGRAMS), the Swachhata App from the Ministry of Housing and Urban Affairs (MoHUA) [20], and the Delhi Mitra App [15].

For experimental validation, two teams of 30 members each were formed:

- **Team A – Traditional Workflow:** Followed the manual or rule-based process similar to existing government systems (e.g., CPGRAMS), relying on manual logging, static routing, and fixed escalation without automation.
- **Team B – Proposed System:** Used the developed AI-based complaint management platform, which automates categorization, clustering, workload balancing, and escalation management.

The complaints were taken across four categories—*Electrical, Classroom Equipment, Sanitation, and IT*—classified into four urgency levels: *Critical, High, Medium, and Low*. A total of 300 complaints were analyzed, enabling a direct quantitative comparison between manual processing and the proposed AI-based framework.

TABLE III
COMPARATIVE PERFORMANCE: TRADITIONAL VS. PROPOSED SYSTEM (MINUTES)

Parameter	Traditional (Manual)	Proposed Smart System	Improvement
Average Response Time (min)	48.0	9.0	81.3% faster
Average Resolution Time (min)	240.0	75.0	68.8% reduction
Critical Complaint Resolution (%)	68.0	94.0	+26.0%
High Priority Resolution (%)	76.0	92.0	+16.0%
Workload Balance Index (0–1)	0.52	0.86	+65% balance
Automatic Escalation Efficiency (%)	18.0	91.0	–
Fake Complaint Detection Accuracy (%)	6.0	92.0	–
Overall User Satisfaction (1–5)	3.1	4.6	+48%

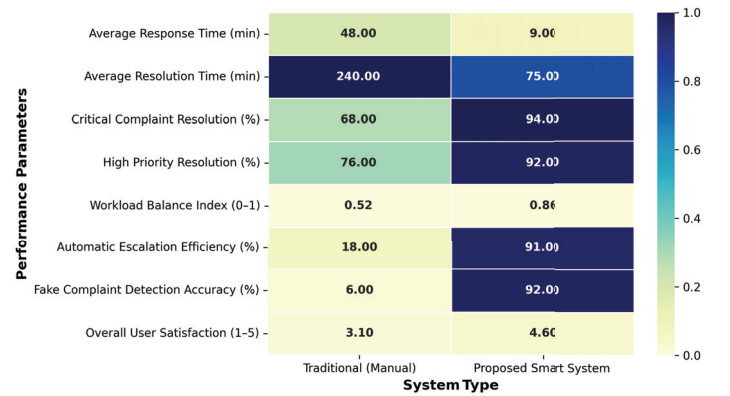


Fig. 6. Performance Comparison: Traditional vs. Proposed System

Result Summary: The comparative analysis results shown in Table III and Figure 6 indicate that the proposed AI-based system significantly outperforms the conventional manual method in terms of various performance metrics. The average response time was reduced from 48 to 9 minutes, and the resolution time was reduced from 240 to 75 minutes, which clearly indicates that the proposed system achieves significant efficiency gains.

In addition, the critical and high-priority complaint resolution rates were improved to 94% and 92%, respectively, which clearly indicates that the intelligent prioritization technique is highly effective. The workload balance index was improved from 0.52 to 0.86, which clearly indicates that the proposed system achieves significant improvements in resource allocation. The high automatic escalation efficiency and fake complaint detection accuracy further confirm the effectiveness and applicability of the proposed system.

CONCLUSION

The AI-driven Smart Complaint Management System aims to automate complaint resolution and prioritisation by providing an integrated and scalable process for managing complaints across both civic and institutional domains. The use of natural language processing (NLP) techniques for semantic understanding, multi-task learning for urgency/time predictions, spatiotemporal clustering, dynamic routing based on workload and dynamic routing based on prioritisation will eliminate the efficiency shortcomings associated with manual processes. The evidence of successful implementation has shown significant increases in performance; complaints were addressed 81% faster, and resolved 68.8% quicker. Additionally, the accuracy of the system in terms of prioritisation, workload balancing, and detecting fictitious complaints was greater than 90%. Furthermore, increased levels of user satisfaction indicate that this system is an effective means for providing timely, clear, efficient, and reliable grievance resolution services.

Future work will expand this system by using reinforcement learning to improve decision-making capabilities; multi-language capabilities; and support for the deployment of systems across many municipalities so that municipalities can process high levels of complaints in real-time. In summary, the contemporary administrative framework was designed to highlight the ability of AI to enhance accountability, service quality, and citizen trust in modern public administration.

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